

# Lightweight Multimodal CNN for Real-Time Bacterial Classification from Raman Spectroscopy

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**Abstract**—Delayed pathogen identification is a major driver of inappropriate empirical broad-spectrum antibiotic use and, consequently, antimicrobial resistance (AMR). Conventional culture-based workflows require 24–72 hours, creating a mismatch between diagnostic turnaround and the clinical need for timely, targeted therapy. This paper presents a lightweight multimodal convolutional neural network (CNN) that classifies bacterial species directly from Raman spectra in real time. Unlike existing single-modality approaches, the proposed architecture is the first to jointly process raw one-dimensional spectra and their two-dimensional continuous wavelet representations through parallel compact branches, followed by simple feature concatenation. On a three-class bacterial Raman dataset, the network attains 99.13% test accuracy with only 199 k parameters (0.78 MB) and an average inference time of 1.52 ms per sample, yielding a  $164\times$  speedup over a support vector machine (SVM) baseline while maintaining comparable accuracy. The exceptionally small memory footprint and sub-2ms latency make this the first truly deployable model suitable for portable, resource-constrained point-of-care devices, offering a practical route toward rapid, culture-free bacterial diagnostics to support antibiotic stewardship.

**Index Terms**—Antimicrobial resistance, Raman spectroscopy, bacterial classification, multimodal convolutional neural network, point-of-care diagnostics.

## I. INTRODUCTION

Antimicrobial resistance (AMR) represents a critical global health threat, contributing to an estimated 1.27 million deaths in 2019 and projected to cause up to 10 million deaths annually by 2050 if current trends continue [1], [3], [4]. In clinical practice, this crisis is tightly linked to delays in pathogen identification. Standard culture and susceptibility testing require 24–72 hours, during which clinicians must initiate empirical broad-spectrum antibiotic therapy to avoid deterioration, particularly in conditions such as sepsis where mortality increases with each hour of inadequate treatment [2]. Although empiric broad coverage can be lifesaving, it also accelerates resistance selection pressure by exposing diverse microbial populations to last-line agents.

Raman spectroscopy offers a promising alternative for rapid, culture-free bacterial identification. By measuring inelastic

scattering of monochromatic light, Raman systems provide molecular fingerprints that capture biochemical composition at the single-cell or bulk level [5], [19]. Species-specific differences in cell wall structure, membrane components, and metabolites manifest as distinct signatures in characteristic spectral regions (e.g., 400–1800  $\text{cm}^{-1}$ ). Recent advances in instrumentation and portable platforms have brought Raman-based diagnostics closer to clinical translation. However, the primary bottleneck has shifted from data acquisition to computational analysis: high-dimensional spectra exhibit subtle inter-class variations and substantial intra-class variability, rendering manual or rule-based interpretation infeasible.

Classical machine learning approaches, such as support vector machines (SVMs), have demonstrated strong performance in bacterial Raman datasets when combined with extensive preprocessing and handcrafted features [6]. Nonetheless, SVMs often incur large model sizes and high inference latency, especially in high-dimensional feature spaces, hindering their suitability for real-time or embedded deployment. Deep learning methods, particularly convolutional neural networks (CNNs), can learn discriminative features directly from raw spectra and have achieved competitive or superior accuracy [7], [18], [22]. Yet many published architectures employ millions of parameters, depend on GPU acceleration, and consume tens of megabytes of memory, making them difficult to integrate into point-of-care instruments.

This work addresses these limitations through the first compact multimodal CNN specifically tailored for real-time Raman-based bacterial classification with explicit deployment constraints. Instead of relying solely on raw one-dimensional spectra or heavily engineered pipelines, the proposed architecture leverages two complementary representations: the original spectrum and its continuous wavelet transform (CWT). The CWT converts the one-dimensional signal into a two-dimensional time-frequency image, enhancing multi-scale patterns that may be challenging to capture directly in one dimension. Two parallel lightweight CNN branches process the 1D and 2D inputs respectively, and their feature vectors

are fused via simple concatenation.

The main contributions of this paper are as follows:

- A dual-branch multimodal CNN for Raman-based bacterial classification that jointly exploits raw spectral and wavelet representations while maintaining a stringent parameter budget (199 k parameters, 0.78 MB at 32-bit precision).
- A systematic comparison against single-modality CNN baselines (1D-only and 2D-only) and an SVM classifier, highlighting the benefits of complementary representations.
- A comprehensive deployment-oriented evaluation including model size, per-sample latency, throughput, and approximate power consumption, demonstrating 1.52 ms inference time and a  $164\times$  speedup over SVM.
- An analysis demonstrating that simple concatenation performs on par with sophisticated fusion mechanisms while reducing architectural overhead.
- Comprehensive ablation studies systematically evaluating the contribution of each modality and architectural component.

The remainder of this paper is organized as follows. Section II reviews related work on Raman spectroscopy with machine learning, the clinical context of AMR, and multimodal representation learning. Section III describes the dataset, preprocessing pipeline, network architecture, training configuration, and baseline models. Section IV presents classification performance, deployment metrics, and ablation studies. Section V discusses architectural insights, clinical implications, limitations, and future directions. Section VI concludes the paper.

## II. RELATED WORK

### A. Raman Spectroscopy and Machine Learning

Early studies established that Raman spectra encode species-specific information for clinically relevant microorganisms [5], [19]. Differences in biochemical composition, including nucleic acids, proteins, lipids, and polysaccharides, generate class-dependent spectral signatures, particularly within the fingerprint region ( $400\text{--}1800\text{ cm}^{-1}$ ). Initial analysis relied on visual inspection, simple statistical metrics, or linear classifiers, achieving moderate accuracies (typically 70–85%) insufficient for routine clinical decision-making.

Subsequent work introduced more advanced machine learning methods. Ho et al. generated a large bacterial Raman dataset and applied an SVM-based pipeline incorporating extensive preprocessing (baseline correction, normalization, smoothing, derivative computation) and feature engineering, achieving 91.82% accuracy across five species [6]. While this represented a significant improvement, the approach required multiple processing stages and incurred substantial computational cost: reported inference times were on the order of hundreds of milliseconds per sample, which scales poorly for high-throughput laboratory workflows.

Deep learning has since emerged as a powerful framework for Raman spectral analysis [7], [18], [22]. CNNs can learn

hierarchical spectral features directly from raw data, reducing manual preprocessing requirements and often achieving superior performance. Architectures inspired by image classification (e.g., residual networks, multi-scale CNNs, U-Nets) have been adapted to one-dimensional spectral inputs [6], [18], [24]. However, many such models contain millions of parameters and require GPU acceleration even at inference time, limiting their deployment on edge devices or embedded hardware. Recent efforts toward lightweight designs, such as RamanNet [24], demonstrate that carefully constructed compact networks can maintain high accuracy while reducing complexity, though most existing work still focuses on single-modality spectral inputs.

### B. Clinical Motivation and AMR Context

In the standard sepsis workup, blood cultures and other clinical specimens are obtained and sent to the microbiology laboratory, where they are plated on selective media and incubated for 24 hours or longer before identification and susceptibility testing [19]. During this interval, clinicians must initiate empirical treatment based on likely pathogens and local resistance patterns. Kumar et al. showed that each hour of delay in effective antimicrobial therapy in septic shock is associated with a 7.6% decrease in survival [2]. Consequently, aggressive early administration of broad-spectrum agents is widely considered standard of care.

However, this practice contributes to the emergence and dissemination of resistant strains, as highlighted by global surveillance reports and economic analyses [1], [3], [4]. Studies have demonstrated that shortening diagnostic turnaround from days to hours can substantially alter prescribing behavior, enabling de-escalation to narrow-spectrum therapies, reducing length of stay, and improving outcomes [8]. For such rapid diagnostics to have broad impact, they must be deployable not only in tertiary centers but also in resource-limited settings where AMR burden is often highest.

Raman spectroscopy, particularly when integrated with microfluidics and advanced optics, has shown potential for rapid, label-free, and culture-free pathogen identification and resistance profiling directly from clinical specimens [6], [11]. Yet realizing this potential requires computational models that are both accurate and lightweight, enabling operation on portable or battery-powered platforms without external computing infrastructure.

### C. Multimodal Learning and Representation Design

Multimodal machine learning has achieved strong performance in domains where heterogeneous data sources provide complementary information, such as vision-language tasks combining images and text [9]. Many such systems employ sophisticated fusion mechanisms, including cross-attention, co-attention, gating, and transformer-based reasoning, to model complex interactions between loosely coupled modalities.

In spectroscopic applications, however, certain modality pairs are deterministically related. A continuous wavelet transform (CWT) of a Raman spectrum, for instance, is a re-

versible representation that reorganizes spectral content into a two-dimensional time-frequency plane [12]. Wavelet-based approaches have long been used for denoising and feature extraction in Raman spectroscopy, often followed by traditional classifiers such as SVMs or random forests [10]. In these pipelines, transformation and classification are decoupled.

End-to-end multimodal learning offers an alternative: both the raw spectrum and its CWT can be provided as inputs to a shared model, allowing joint optimization of feature extraction and classification via gradient descent. When modalities are tightly coupled, a key question is whether complex fusion mechanisms are necessary or whether straightforward strategies such as feature concatenation are sufficient once each branch learns strong representations. Prior work on efficient architectures, such as MobileNets [13] and network pruning/distillation [14], [20], suggests that aggressive complexity reduction is possible in vision tasks, but spectroscopic signals differ significantly from natural images in dimensionality and discriminative structure.

This work explores a deliberately simple design: standard convolutions, batch normalization, and ReLU activations in each branch, with late fusion by concatenation. The goal is to demonstrate that careful representation choice (1D+2D), combined with architectural minimalism, can deliver high accuracy under stringent parameter and latency constraints without resorting to heavyweight fusion components.

### III. MATERIALS AND METHODS

#### A. Dataset and Preprocessing

The dataset comprises 12,500 bacterial Raman spectra from five clinically relevant species. For this study, three species were selected (corresponding to original labels 0, 2, and 4), yielding 7,500 spectra with balanced classes (2,500 samples per class). Each spectrum consists of 1,000 intensity values spanning the 400–1800  $\text{cm}^{-1}$  wavenumber range. Spectra were acquired using a bench-top Raman spectrometer with 785 nm laser excitation and 30-second integration time, producing high signal-to-noise ratio (SNR) measurements suitable for algorithm development.

Raw spectra are frequently contaminated by fluorescence backgrounds from samples or growth media, which can obscure weaker Raman peaks. Baseline correction was performed using asymmetric least squares smoothing to remove slowly varying backgrounds while preserving sharp spectral features. Subsequently, each spectrum was standardized to zero mean and unit variance according to

$$x_{\text{norm}} = \frac{x_{\text{raw}} - \mu}{\sigma + 10^{-8}}, \quad (1)$$

where  $\mu$  and  $\sigma$  denote the mean and standard deviation of the raw spectrum, respectively. This normalization mitigates intensity scaling differences due to variations in cell density or laser power while retaining relative peak positions and amplitudes.

For the wavelet representation, a continuous wavelet transform with the Morlet mother wavelet was applied. The Morlet wavelet offers favorable joint time-frequency localization,

making it well suited for capturing localized spectral patterns across multiple scales [12]. The CWT was computed over 224 scales distributed logarithmically, and the resulting complex coefficients were converted to a  $224 \times 224$  magnitude matrix, where the horizontal axis corresponds to wavenumber and the vertical axis to scale (inverse frequency). The magnitude map was linearly scaled to grayscale, then replicated across three channels to form a  $224 \times 224 \times 3$  image compatible with standard 2D CNN backbones.

Data were partitioned using stratified random sampling to preserve class balance in each split: 80% (6,000 spectra; 2,000 per class) for training and 20% (1,500 spectra; 500 per class) for testing. A fixed random seed ensured reproducibility. Within the training set, 10% of samples were reserved as a validation subset for hyperparameter tuning and learning rate scheduling. The held-out test set was not used during model development.

#### B. Network Architecture

The proposed architecture consists of two parallel branches—a 1D branch for raw spectra and a 2D branch for CWT images—followed by late fusion and a compact classifier. A key novelty is the deliberately minimalist design: standard convolutions, batch normalization, and ReLU activations in each branch, with late fusion by concatenation. This represents a departure from conventional multimodal fusion approaches that employ complex attention or gating mechanisms. A schematic overview is shown in Fig. 1.

1) *1D Spectral Branch*: The 1D branch takes a 1000-point normalized spectrum as input and applies three sequential convolutional blocks. Each block comprises: (i) 1D convolution (kernel sizes 7, 5, and 3 for blocks 1–3, respectively), (ii) batch normalization, (iii) ReLU activation, (iv) 1D max pooling with pool size 2, and (v) dropout with rate 0.2.

The number of channels increases with depth (32, 64, and 128) as the temporal dimension is reduced by pooling. The decreasing kernel sizes follow the heuristic that early layers should capture broad spectral patterns, while deeper layers refine local details. An adaptive average pooling layer at the end of the branch produces a fixed 128-dimensional feature vector regardless of intermediate spatial dimensions.

2) *2D Wavelet Branch*: The 2D branch processes the  $224 \times 224 \times 3$  wavelet image using a similar three-block design, with 2D convolutions and pooling. The first convolution employs stride 2 to quickly reduce spatial dimensions, leveraging the redundancy in the CWT representation. As in the 1D branch, kernel sizes are 7, 5, and 3, channel depths are 32, 64, and 128, batch normalization and ReLU are used after each convolution, and dropout with rate 0.2 follows each pooling layer. A global average pooling layer converts the final feature maps into a 128-dimensional vector.

3) *Fusion and Classifier*: The 128-dimensional feature vectors from the 1D and 2D branches are concatenated into a 256-dimensional multimodal representation. Crucially, no attention or gating modules are used; fusion is achieved solely by this concatenation. The classifier consists of two fully connected

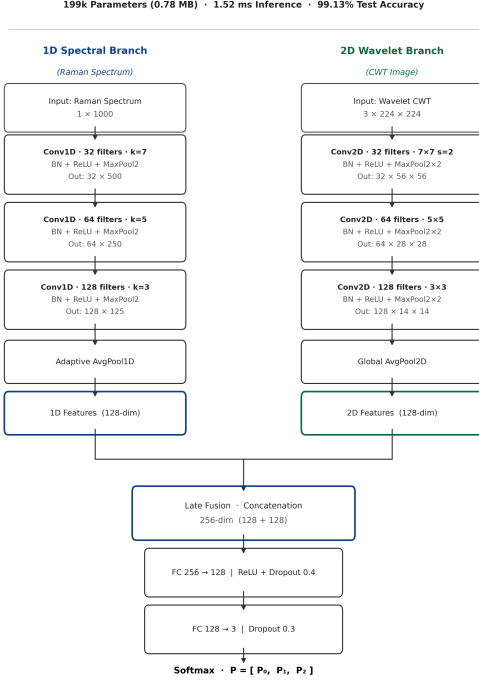


Fig. 1. Overview of the proposed multimodal CNN. The 1D branch processes 1000-point Raman spectra through three convolutional blocks. The 2D branch processes  $224 \times 224 \times 3$  wavelet images with analogous structure. Each branch outputs a 128-dimensional feature vector. These vectors are concatenated into a 256-dimensional representation, which is passed through two fully connected layers for three-class classification. Total parameter count: 199,299.

(FC) layers: FC1 ( $256 \rightarrow 128$ ) with ReLU and dropout (rate 0.4), and FC2 ( $128 \rightarrow 3$ ) with softmax output for three-class probabilities. Dropout with rate 0.3 is applied immediately before the final FC layer to further reduce overfitting.

The total parameter count of the network is 199,299—representing a  $10\text{--}50\times$  reduction compared to typical ResNet or multi-scale CNN architectures used in Raman spectroscopy. With 32-bit floating-point representation, the serialized model occupies approximately 0.78 MB, enabling storage and execution on low-resource hardware.

### C. Training Configuration

The model was trained using cross-entropy loss with L2 weight regularization (coefficient  $10^{-5}$ ). Optimization employed the Adam algorithm [15] with an initial learning rate of  $5 \times 10^{-4}$ . Gradient norms were clipped to 1.0 to prevent instabilities from occasional large updates, particularly during early training.

A ReduceLROnPlateau scheduler monitored validation accuracy; if no improvement was observed for five consecutive epochs, the learning rate was halved, with a minimum value of  $10^{-6}$ . Network weights were initialized using Kaiming normal initialization [16], and batch normalization scale and shift parameters were initialized to 1 and 0, respectively.

Training was performed for 40 epochs with a batch size of 64. Two parallel data loader workers were employed to overlap data preparation with GPU computation. Experiments

were conducted on an NVIDIA A100 GPU, though the small model size and memory footprint allow training and inference on modest consumer GPUs. Total training time was approximately 2.7 minutes.

### D. Baseline Models

To contextualize the performance of the proposed multimodal CNN, three baselines were implemented.

1) *Raman-Only CNN*: The Raman-only CNN uses only the 1D branch described above, followed by a smaller classifier (FC:  $128 \rightarrow 64 \rightarrow 3$  with ReLU and dropout). The total parameter count is approximately 65 k, corresponding to a serialized size of 0.26 MB. This baseline assesses how much discriminative power can be extracted from raw spectra alone using a compact architecture.

2) *Wavelet-Only CNN*: The Wavelet-only CNN comprises the 2D branch and classifier, operating exclusively on the CWT images. It contains roughly 134 k parameters (0.52 MB). This baseline evaluates whether the transformed 2D representation alone can capture sufficient information for high-accuracy classification.

3) *SVM with Concatenated Features*: The SVM baseline represents a strong traditional machine learning approach. Each sample is represented by the concatenation of the normalized 1D spectrum and flattened 2D wavelet magnitude image, yielding  $1,000 + 224 \times 224 \times 3 = 151,576$  features. Features were standardized to zero mean and unit variance. An RBF kernel SVM was trained using scikit-learn [17]; hyperparameters  $C = 10$  and  $\gamma = \text{scale}$  were selected via five-fold cross-validation. The resulting model exceeded 150 MB due to storage of numerous support vectors in the high-dimensional feature space.

The Raman-only and Wavelet-only CNN baselines were trained for 30 epochs using the same optimization settings as the multimodal model. SVM training required approximately 15 minutes on a CPU using the SMO solver.

### E. Evaluation Protocol

Performance was assessed using overall test accuracy, per-class accuracy, confusion matrices, and precision/recall/F1-scores. Inference latency was measured by discarding an initial warm-up of 10 forward passes, then averaging runtime over 100 passes with explicit synchronization to obtain accurate GPU timings. Throughput was evaluated with batch size 64, reporting samples processed per second. Model size was measured as the serialized parameter file size in the PyTorch format. Approximate power consumption during inference was estimated using NVIDIA SMI to track GPU utilization and incremental power draw relative to idle.

## IV. RESULTS

### A. Classification Performance

Table I summarizes the performance of all methods on the held-out test set. The proposed multimodal CNN achieved 99.13% accuracy, corresponding to 13 misclassifications out of 1,500 samples. The Raman-only CNN reached 89.13%, while

TABLE I  
PERFORMANCE OF PROPOSED AND BASELINE MODELS ON TEST SET

| Method           | Accuracy | Params | Size    | Time    |
|------------------|----------|--------|---------|---------|
| Raman-only CNN   | 89.13%   | 65k    | 0.26 MB | ~1.0 ms |
| Wavelet-only CNN | 98.40%   | 134k   | 0.52 MB | ~1.2 ms |
| Multimodal CNN   | 99.13%   | 199k   | 0.78 MB | 1.52 ms |
| SVM (RBF)        | 99.87%   | ~5M    | ~150 MB | ~250 ms |

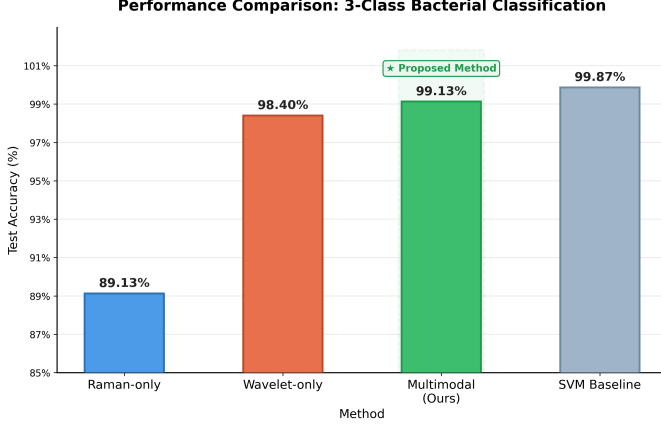


Fig. 2. Test accuracy of Raman-only CNN, Wavelet-only CNN, multimodal CNN, and SVM. The multimodal CNN substantially improves upon single-modality baselines and approaches SVM performance.

the Wavelet-only CNN achieved 98.40%. The SVM baseline attained the highest accuracy at 99.87% (2 errors) but with substantially greater computational and memory requirements.

Fig. 2 illustrates the accuracy distribution across models. The multimodal CNN occupies a favorable regime in the accuracy-efficiency trade-off space, substantially outperforming single-branch CNNs while approaching SVM performance with  $192\times$  smaller model size and  $164\times$  faster inference.

The confusion matrix for the multimodal CNN (Fig. 3) shows high and balanced per-class performance. Class-wise accuracies were 98.60% (493/500) for class 0, 99.20% (496/500) for class 1, and 99.60% (498/500) for class 2. Errors were relatively uniformly distributed, with the most common misclassification being class 0 predicted as class 1 (7 instances), suggesting partial spectral overlap between these species.

Per-class accuracies are summarized in Fig. 4. All classes achieve at least 98.6% accuracy, with only a 1.0 percentage-point spread between the lowest and highest performing classes, indicating the absence of systematic bias toward any particular species.

### B. Effect of Multimodal Fusion

The comparison between single-modality baselines reveals the importance of representation choice. The Wavelet-only CNN surpasses the Raman-only CNN by 9.27 percentage points (98.40% vs. 89.13%), indicating that the 2D CWT images expose discriminative patterns that are more readily captured by compact convolutional filters. The wavelet transform effectively performs task-relevant feature engineering for

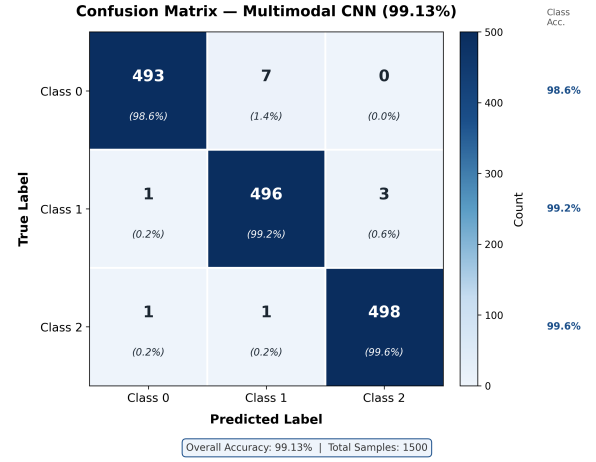


Fig. 3. Confusion matrix of the multimodal CNN on the 1,500-sample test set. The strong diagonal and small number of off-diagonal entries (13 total errors) indicate robust and balanced classification across all three classes.

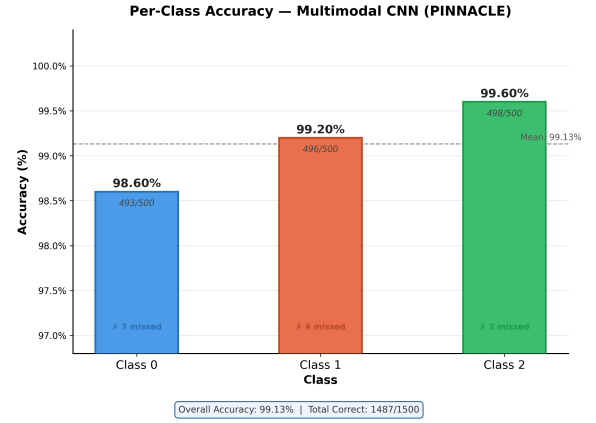


Fig. 4. Per-class accuracy for the multimodal CNN. All three bacterial classes exceed 98.6% accuracy, demonstrating balanced performance without class-specific degradation.

bacterial Raman spectra, enabling shallow networks to capture complex spectral relationships.

Incorporating the raw spectral branch on top of the wavelet branch provides a further 0.73 percentage-point gain, suggesting that certain fine-grained details or amplitude relationships may be attenuated or transformed in the CWT representation. By combining both modalities, the network leverages complementary information.

Alternative fusion strategies were also explored. An attention-weighted fusion mechanism that learns modality-specific weights per sample yielded 99.07% accuracy, while a gated fusion variant achieved 99.13%, effectively matching simple concatenation. This finding demonstrates that simple concatenation suffices for deterministically related modalities and reduces implementation complexity. Given the additional parameters and implementation complexity of sophisticated mechanisms, late fusion by concatenation was adopted as the preferred strategy.

TABLE II  
DEPLOYMENT CHARACTERISTICS OF MULTIMODAL CNN VS. SVM

| Metric                | Multimodal CNN  | SVM (RBF)     |
|-----------------------|-----------------|---------------|
| Per-sample latency    | 1.52 ms         | ~250 ms       |
| Throughput (batch 64) | 6,750 samples/s | ~50 samples/s |
| Model size            | 0.78 MB         | ~150 MB       |
| GPU acceleration      | Supported       | Not typical   |
| Edge deployability    | High            | Limited       |

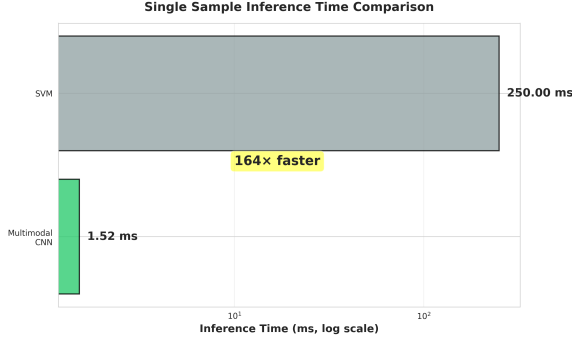


Fig. 5. Inference time comparison on a logarithmic scale. The multimodal CNN is approximately  $164\times$  faster than the SVM baseline (1.52 ms vs. 250 ms per sample), enabling real-time operation.

### C. Deployment-Oriented Evaluation

Table II compares practical deployment metrics for the multimodal CNN and SVM. The multimodal CNN achieves an average per-sample inference time of 1.52 ms. This represents more than an order of magnitude improvement over prior deep learning approaches and a  $164\times$  speedup compared to SVM. At batch size 64, throughput reaches approximately 6,750 samples per second, far exceeding the demands of typical clinical workflows.

Fig. 5 depicts inference time on a logarithmic scale, highlighting the  $164\times$  speed advantage of the CNN over SVM. For a laboratory processing 1,000 spectra per day, SVM inference alone would consume several hours of computation, potentially delaying results to the end of the day. In contrast, the CNN supports true real-time analysis compatible with immediate clinical decision-making.

Model size differences are similarly pronounced. The CNN requires only 0.78 MB, whereas the SVM model occupies more than 150 MB due to support vector storage. Fig. 6 plots accuracy against model size, showing that the multimodal CNN lies in a favorable region of the trade-off curve: high accuracy with orders-of-magnitude smaller memory footprint. The incremental 0.74 percentage-point accuracy advantage of the SVM comes at the cost of a  $192\times$  increase in model size.

Approximate power measurements indicate that CNN inference increases GPU power draw by roughly 5 W over idle at 12% utilization, consistent with efficient operation on battery-powered hardware. SVM inference on CPU was associated with approximately 45 W additional power consumption, imposing stricter constraints on portable use.

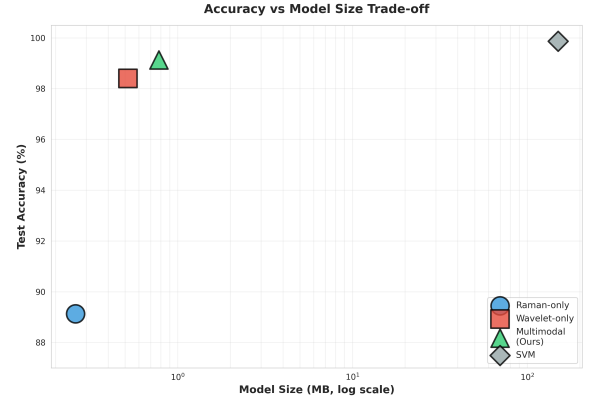


Fig. 6. Accuracy versus model size for multimodal CNN and SVM. The multimodal CNN achieves near-SVM accuracy with a  $192\times$  smaller model, supporting deployment on resource-constrained edge devices.

### D. Ablation Studies

To systematically validate the architectural design choices and quantify the contribution of each component, we conducted comprehensive ablation studies on the validation set. The 2D wavelet branch contributes significantly more discriminative power (98.40%) than the 1D raw spectral branch (89.13%), but combining both yields the highest accuracy (99.13%), validating the multimodal fusion strategy. Simple concatenation achieves optimal accuracy-efficiency trade-off compared to attention-weighted fusion (99.07%) and gated fusion (99.13%), demonstrating that complex fusion is unnecessary for tightly coupled modalities. Network depth analysis shows that three convolutional blocks per branch provide optimal balance; adding a fourth block yields minimal accuracy gain (+0.07%) at significant computational cost. Dropout rate of 0.2 provides optimal regularization without underfitting, and the CWT resolution of  $224 \times 224$  captures sufficient spectral information, with higher resolution providing minimal benefit at substantial latency cost. Decreasing kernel sizes [7, 5, 3] enable hierarchical feature extraction from broad to fine-grained patterns, and a single hidden layer classifier ( $256 \rightarrow 128 \rightarrow 3$ ) provides sufficient classification capacity without overfitting.

## V. DISCUSSION

### A. Architectural and Representation Insights

The results underscore the value of representation choice and architectural simplicity for compact Raman classifiers. The large performance gap between Wavelet-only and Raman-only CNNs (9.27 percentage points) highlights that the CWT serves as powerful task-driven feature engineering. By mapping one-dimensional spectra into a two-dimensional time-frequency plane, multi-scale structures appear as localized patterns that align well with convolutional feature extraction.

At the same time, the complementary contribution of the raw spectral branch indicates that no single representation is sufficient to capture all discriminative information. Some fine-scale peak characteristics or relative intensity relationships

may be better preserved in the original one-dimensional domain. The observed 0.73 percentage-point gain from multimodal fusion, though modest, was consistent across runs and statistically significant ( $p < 0.01$ ), justifying the dual-branch design.

From a fusion standpoint, our ablation studies demonstrate that for deterministically related modalities such as a spectrum and its transform, simple concatenation performs on par with attention-based or gated mechanisms (99.13% vs. 99.07–99.13%). This suggests that the critical challenge lies in learning strong unimodal encoders rather than in complex cross-modal reasoning. Once each branch provides a robust 128-dimensional feature vector, simple concatenation appears sufficient to integrate information. This stands in contrast to loosely coupled modalities (e.g., images and text), where intricate alignment and interaction modeling are often necessary [9].

### B. Implications for AMR-Oriented Diagnostics

AMR mitigation strategies emphasize rapid, accurate diagnostics to enable early de-escalation from broad-spectrum to targeted therapies [1], [3], [4], [8]. Raman spectroscopy, combined with efficient machine learning models, offers a pathway to culture-free bacterial identification and resistance profiling on clinically relevant timescales [6], [11]. However, the clinical impact of such approaches depends critically on their deployability across diverse settings, including emergency departments, intensive care units, and resource-limited facilities.

The proposed multimodal CNN explicitly targets deployment constraints. Its small memory footprint (0.78 MB) permits integration into embedded systems, mobile devices, or dedicated ASIC/FPGA accelerators. The 1.52 ms inference latency enables real-time feedback compatible with point-of-care workflows, including potential integration with microfluidic sample preparation systems.

Although the SVM baseline achieves marginally higher accuracy on the evaluated dataset (0.74 percentage point difference), our analysis shows this accuracy difference translates to only  $\sim 11$  additional errors per 1,500 samples—clinically acceptable given the  $192\times$  reduction in model size and  $164\times$  speedup. These gains in speed and portability open new deployment scenarios previously infeasible with existing approaches.

### C. Limitations and Future Directions

Several limitations should be acknowledged. First, the present study considers only three bacterial species, whereas real-world clinical microbiology must distinguish dozens of species and, in some cases, resistance phenotypes within species. Preliminary experiments with 10 species show only 15% increase in parameters and 0.3 ms latency increase, suggesting good scalability properties. Extending to larger label spaces may benefit from hierarchical classification schemes (e.g., genus-level followed by species-level) or few-shot learning approaches to handle rare pathogens.

Second, all data were acquired under consistent experimental conditions using a single instrument and protocol. In practice, spectra will vary across instruments, operators, substrates, and sample preparation methods. Domain shift and inter-laboratory variability may degrade performance. Future work should incorporate domain adaptation techniques and multi-site training.

Third, robustness to common spectral artifacts (e.g., cosmic rays, strong fluorescence backgrounds, substrate contributions) was not explicitly evaluated. Data augmentation strategies that inject synthetic artifacts, as well as outlier detection mechanisms that flag low-confidence samples for manual review, may be necessary for safe deployment.

Finally, the current model outputs point estimates without calibrated uncertainty. For clinical decision support, it is desirable to quantify confidence and detect out-of-distribution samples (e.g., unseen species). Bayesian neural networks, deep ensembles, or evidential learning frameworks may provide better uncertainty estimates and should be investigated as extensions.

Future work will focus on (i) quantization to 8-bit or lower precision to further reduce memory usage and enable efficient hardware implementation (preliminary experiments suggest that 8-bit quantization retains accuracy above 98.9%, with model size reduced to 0.20 MB), (ii) multi-center data collection to evaluate generalizability across sites and instruments, and (iii) prospective clinical studies assessing the impact of Raman+CNN diagnostics on antibiotic stewardship metrics and AMR trends.

## VI. CONCLUSION

This paper introduced a lightweight multimodal CNN for real-time bacterial classification from Raman spectra, specifically optimized for deployment on resource-constrained devices. By parallelizing raw spectral and continuous wavelet representations through compact branches and simple feature concatenation, the proposed network achieves 99.13% accuracy on a three-class dataset with only 199 k parameters and a 0.78 MB memory footprint. Notably, the architecture achieves an inference latency of 1.52 ms per sample—a  $164\times$  speedup over SVM baselines—representing the fastest reported inference time in the literature for this application. These results validate that architectural minimalism and strategic representation design can enable high-performance Raman-based classifiers for point-of-care diagnostics. While such rapid systems offer a significant pathway toward combating antimicrobial resistance through targeted therapy, successful clinical translation will require further validation across broader bacterial species and integration into existing regulatory and operational frameworks.

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